



Potato: Crop Stage-wise IPM

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Management

Activity

Pre sowing*

Common cultural practices:

- Summer deep ploughing
- Soil solarization during summer.
- Field sanitation, rogueing.
- Avoid water logged conditions in the field.
- Follow crop rotation.
- Apply manures and fertilizers as per soil test recommendations
- Start to grow ecological engineering plants.
- Sow/plant 4 rows of maize, sorghum, bajra (pearl millet) around the potato crop field as a guard/barrier crop.

Nutrient

- Apply FYM @ 8 t/acre or vermicompost @ 4-6 t/acre
- Apply 2 Kg each of Azospirillum and Phosphobacterium with 10 Kg FYM /acre as soil application before planting

Weed

- Stale seed bed technique before sowing.
- Destroy all the germinated weeds by shallow ploughing before sowing.

Cultural control:

- Give light irrigation and cover the beds with polythene

Soil borne pathogens, nematodes and resting stages of insect pests

- Give light irrigation and cover the beds with polythene sheet of 45 gauge (0.45 mm) thickness for three weeks before sowing.
- Raise African marigold in the nursery 15 days prior to sowing against cyst nematode.
- Use raised seed beds of more than 35cm height (for better water drainage).

Biological control:

- Apply neem cake@ 80 Kg/acre.

Sowing*

Common cultural practices:

- Use resistant/tolerant varieties.
- Use healthy, certified and weed seed free tubers.

Nutrients

- Apply 48 Kg nitrogen (N), 16 Kg phosphorus (P_2O_5) and 48 Kg potassium (K_2O)/acre.
- Apply N and K in two splits; half as basal and half as top dressing at 30 days after sowing.

Plant growth regulator (PGR)

- Dip cut pieces of tuber (seed) for 10 minutes in chlormequat chloride 50% SL @ 100 ppm solution.

Weeds

- Adopt recommended agronomic practices like field preparation, time of sowing, row and plant spacing, gap filling etc. to obtain the healthy plant stand to reduce the weed menace.
- If weed flora of the field is known based on previous season experience the preemergence recommended herbicide oxyflourfen 23.5% EC @170-340 ml in 200-300 l water/ acre be applied within 3-4 days after sowing.
- When 5-10 % tubers are germinated, application of paraquat dichloride 24 % SL @ 200 gram a.i./acre can be used to control broadleaf, sedges and grassy weeds.

Cultural control:

- Tubers stored in oxygen deficient structures should not be used.

Chemical control:

Resting stage of the diseases, black heart/ scurf

- Treat tuber with M.E.M.C. 6% FS @ 0.415 g/ Kg tubers in 100 ml water for 3-5 minute or soaking potato seed tubers in streptocycline 40 to 100 ppm solution for half an hour prior to planting or with carbendazim 25%+ mancozeb 50% WS @ (1.5 + 3.0) to (1.75 + 3.5) for 10 Kg seed (tuber) or with carboxin 37.5% + thiram 37.5% DS@ (1.5 + 3.0) to (1.75 + 3.5) for 10 Kg seed (tuber).

Vegetative/ seed/tuber stage

Common cultural practices:

- Collect and destroy crop debris
- Judicious use of fertilizers
- Provide irrigation at critical stages of the crop
- Avoid water logging
- Avoid any stress to the crop as much as possible
- Enhance biocontrol activity by avoiding chemical spray, when 1-2 natural enemies are observed.

Common mechanical practices:

- Collect and destroy disease infected and insect infested plant parts
- Collect and destroy eggs and early stage larvae
- Handpick the older larvae during early stages of crop
- Use yellow and blue sticky traps @ 4-5 trap/acre
- Use light trap @ 1/acre and operate between 6 pm and 10 pm
- Install pheromone traps @ 4-5/acre for monitoring adult moths activity (replace the lures with fresh lures after every 2-3 weeks)
- Erect bird perches @ 20/acre for encouraging predatory birds such as King crow, common mynah etc.
- Set up bonfire during evening hours at 7-8 pm

Common biological practices:

- Conserve natural enemies through ecological engineering
- Augmentative release of natural enemies

Cultural control:

- Use pathogen free tubers

Bacterial wilt

- Use pathogen free tubers.
- Disinfect the cutting knife using 1% sodium hypochlorite solution.
- Apply lime (dolomite) in the soil as acidic or alkaline soil is not conducive to the bacterial wilt pathogen.

Biological control:

- Apply neem cake @ 80 Kg/acre

Chemical control:

- Two to three sprays of (streptomycin sulphate 9% + tetracylin hydrochloride 1%) SP @ 40 to 50 ppm solution at an interval of 20 days. First spray 30 days after planting.

Cultural control:

Root-knot nematode

- Plant crop during the 2nd week of October in autumn and in early January to limit rootknot nematode infestation on tubers.
- Grow one row of repellent plants like Tagetes patula and T. erecta (African marigold) in between 2 or 3 rows of potatoes.

Biological control:

- Apply NSKE 4% and neem cake @ 80 Kg/acre.

White grub / root grub

Follow common cultural, mechanical and biological practices

Biological control:

- Entomopathogenic nematodes (EPNs) can be sprayed at the rate of 1 billion nematodes per acre, in white grub / root grub infested fields OR
- EPN infected cadavers of Galleria/Corcyra larvae containing live infective juveniles (IJs) are implanted in soil at plant bases at the rate of four cadavers per plant during May/June and/or September for white grub / root grub control.

Tuber initiation

Leaf spot complex

- Follow common cultural, mechanical and biological practices

Early blight

- Follow common cultural, mechanical and biological practices

Chemical control:

- Spray aureofungin 46.15% w/v. SP @ 0.005% in 300 l of water/acre or captan 50% WG @ 600 g in 200 l of water/acre (second spray after 5 days interval) or captan 50% WP @ 1 Kg in 300- 400 l of water/acre or captan 75% WP @ 666 g in 400 l of water/acre. (second spray after 8 days interval) or chlorothalonil 75% WP @ 350-500 g 240-320 l of water/acre (second spray after 14 days interval) or copper oxychloride 50% WP @ 1 Kg in 300-400 l of water/acre or mancozeb 35% SC @ 0.5% or 500 g/100 l water 500 l water or as required depending upon crop stage and equipment used or mancozeb 75% WP@ 600-800 g in 300 l of water/acre or hexaconazole 2% SC @ 1.2 l in 200 l of water/acre (second spray after 21 days interval) or kitazin 48% EC @ 0.20% or 200 ml in 200 l of water or propineb 70% WP @ 300 g in 100 l of water or 0.30% as required depending upon crop stage and plant protection equipment used (second spray after 15 days interval) or zineb 75% WP @ 600- 800 g in 300-400 l of water/acre or captan 70% + hexaconazole 5% WP @ 200- 400 g in 200 l of water/acre (second spray after 21 days interval).

- Follow common cultural, mechanical and biological practices

Cultural control:

- Use short-duration varieties.
- The model specifies that 7 days moving sum of RH > 85% for at least 90 hr coupled with a 7 day moving sum of temperature between 7.2 and 26.6°C for at least 115 hr would predict appearance of late blight within 10 days of satisfying the conditions.

Chemical control:

- Spray captan 50% WG @ 600 g in 200 l water (second spray after 5 days interval) or captan 50% WP @ 1 Kg in 300- 400 l water/acre or captan 75% WP @ 666 g in 400 l water/acre (second spray after 8 days interval) or

Late blight

water/acre (second spray after 8 days interval) or chlorothalonil 75% WP @ 350-500 g in 240-320 l of water/acre (second spray after 14 days interval) or copper oxychloride 50% WP @ 1 Kg in 300-400 l of water/acre or copper sulphate 2.62 % SC @ 400 ml in 200 l of water/acre (second spray after 3 days interval) or cyazafamid 34.5% SC @ 80 ml in 200 l water/acre (second spray after 27 days interval) or dimethomorph 50% WP@ 400 g in 300 l water/acre (second spray after 16 days interval) or mancozeb 75% WG @ 400 in 200 l water/acre (second spray after 3-5 days interval) or mancozeb 75% WP@ 600-800 g in 300 l water/acre or hexaconazole 2% SC @ 1.2 l in 200 l water/acre (second spray after 21 days interval) or mandipropamid 23.4% SC @ 0.2 ml/ l in 200- 300 l of water/acre (second spray after 40 days interval) or propineb 70% WP @ 300 g in 100 l of water or 0.30% as required depending upon crop stage and plant protection equipment used (second spray after 15 days interval) or zineb 75% WP@ 600- 800 g in 300-400 l of water/acre or captan 70% + hexaconazole 5% WP @ 200- 400 g in 200 l of water/acre (second spray after 21 days interval) or cymoxanil 8% + mancozeb 64% WP @ 600- 800 g in 200-300 l of water/acre (second spray after 10 days interval) or famoxadone 16.6% + cymoxanil 22.1% SC @ 200 ml in 200-300 l of water/acre (second spray after 27 days interval) or fenamidone 10% + mancozeb 50% WDG @ 500- 600 g in 200 l of water/acre (second spray after 30 days interval) or metalaxyl M 4% + mancozeb 64% WP @ 0.25% 1 Kg/ acre in 200-400 l water (second spray after 24 days interval) or metalaxyl 8% + mancozeb 64% WP @ 0.25% 1 Kg/ acre in 400 l water (second spray not less than 7 weeks) or metiram 55% + pyraclostrobin 5% WG @ 600-700 g in 200 l water/ acre (second spray after 15 days interval) or azoxystrobin 23% SC@200 ml in 200 l of water/acre or treat tuber with carbendazim 25% + mancozeb 50% WS @ (1.5 + 3.0) to (1.75 + 3.5) for 10 Kg seed (tuber).

- Follow common cultural, mechanical and biological practices

Leaf curl diseases

Cultural control:

- Use peppermint repellent plant for whitefly (vector).
- Show attractant plant like French bean to attract predatory thrips.

Spider mites**

- Follow common cultural, mechanical and biological practices

Cultural control:

- Grow flowering plants for natural enemies attraction: carrot family, bishop's weed (spider mite destroyer), sunflower family, French bean (predatory mites) etc.

Leaf miner**

- Follow common cultural, mechanical and biological practices

Tobacco caterpillar**

- Follow common cultural, mechanical and biological practices

Cultural control:

- Grow castor as ovipositional trap crop.

Biological control:

- Release egg parasitoid, *Trichogramma pretiosum* @ 20,000/acre/week four times.
- Spray NSKE 5% against eggs and first instar larva.
- Apply entomopathogenic nematodes (EPNs) @ 2,50,000 infective juveniles of *Steinernema feltiae*/sq mt area.

Aphid, thrips** and leaf hopper (aphid's transmit potato virus

- Follow common cultural, mechanical and biological practices

Cultural control:

- Use healthy seed, hot and cold weather cultivation, green manuring, irrigation, fertilizer application.
- Plant early bulking and/or maturing cultivars to help seed production programme in areas having short aphid-free periods so that the seed crop may escape the population pressure of aphid vectors.

Biological control:

- Spray NSKE 5%.

Y (PVY) and potato
leaf roll virus (PLRV),
whitefly

Chemical control:

- Apply carbofuran 3% CG @ 6.64 Kg/ acre or oxydemeton-methyl 25% EC @ 0.4 l in 200-400 l of water/acre or thiamethoxam 25% WG @ 40 g in 200 l of water/acre or phorate 10% CG @ 4 Kg/ acre or soil drenching of thiamethoxam 25% WG @ 80 g in 200 l water/acre for aphid control.
- Spray carbofuran 3% CG @ 1.328 Kg/ acre for controlling leaf hopper
- Spray dimethoate 30% EC @ 264 ml in 200- 400 l water/acre for controlling thrips.

Cutworms**

- Follow common cultural, mechanical and biological practices

Cultural control:

- Attracting cutworm larvae using rice bran – heaps of rice bran should be placed in several places in the late afternoon. They can be removed from the rice bran on the next day and destroyed.
- Flood field prior to planting - where/whenever possible farmers can consider temporarily flooding fields, particularly on severely infested fields.

Potato wart

- Follow common cultural, mechanical and biological practices

Cultural control:

- Adopt intercropping potato with maize or rotational crops such as bean and radish (reduce population of viable resting spores in soil)

Potato scab

- Follow common cultural, mechanical and biological practices

Cultural control:

- Use healthy tubers and treat the seed tubers with boric acid (3% for 30 minutes) before or after cold storage.
- Maintain optimum soil moisture from tuber initiation.

- Practice crop rotation with wheat, peas, oats, barley, lupin, soybean, sorghum, bajra and green manures crops.

Chemical control:

- Spray thiram 75% WS @ 25 g/ l water (second spray after 7-10 days interval)

Vegetative and tuber development stage

Nutrients

- Apply 2nd half of N & K at 30 days after planting after hoeing / weeding.

Plant growth regulators

- Apply mepiquat chloride 5% AS @ 1.25-1.5 l (mix 200 -300 ml of products in 10 l of water) at 45 days after sowing to restrict the excessive vegetative growth of potato and increasits yield
- Apply gibberellic acid 0.001% L @ 180 ml in 450-500 l of water twice at 45 days after sowing and 65 days after sowing or Apply triacontanol 0.05% EC @ 0.50 l in 500-600 l of water twice at 30 and 45 days after sowing to enhance the yield

Weeds

- Hand tool weeding/hoeing along with earthing twice at 30 and 60 days after planting.
- Pre-emergence application of oxyflourfen 23.5% EC @170-340 ml in 200-300 l of water/acre
- Post-emergence overall/ inter-row application (at 5-10% emergence) of paraquat dichloride 24% SL @ 800 ml in 200 l of water/acre or 2,4-D dimethyl amine salt 58% SL @1.376 l in 160 l of water/acre.
- Use mulch on ridges to suppress weed growth such as paddy straw, maize or sorghum stalks or farm refuses.
- In hilly regions, used local available materials such as pine needles or leaf litter as

mulch for controlling weeds and reduce run off loss and conserving moisture.

- Follow common cultural, mechanical and biological practices

Cultural control:

- Grow intercrops such as cowpea, onion, maize, coriander, urdbean in 1:2 ratio
- Rotate the potato crop with a non host cereal, cucurbit, or cruciferous vegetable crop
- Grow repellent plants such as ocimum/basil
- Use healthy seed, hot and cold weather cultivation, green manuring, irrigation, fertilizer application, storage and adoption of seed plot technique

Potato tuber moth

- Plant seed tubers at a depth of 10 cm
- The fields should be ridged after 6 to 7 weeks of planting so that the tubers are buried at least 25cm below the soil surface
- Timely and adequate irrigations minimize soil cracking and thereby reduce the risk of tuber exposure to potato tuber moth attack or their egg laying
- Always keep the tubers underneath the soil surface

Biological control:

- Inundative release of *T. pretiosum* @ 0.4 lakh/acre 4-5 times from flower initiation stage at weekly intervals

Bacterial diseases, black surf / canker

- Follow common cultural, mechanical and biological practices

Cultural control:

- Soak seed tubers in a solution of trisodium phosphate (90 g/l of water) one day before sowing. The tubers should be thoroughly rinsed and dried in shade.

Potato virus Y, S, X

- Same as in aphid/thrips control

- Follow common cultural, mechanical and biological practices

Bacterial soft rot

Cultural control:

- Physical damage must be avoided as it encourages post harvest rots.

Harvesting

Weeds

- Prior to harvesting/ after harvesting left over weeds should be removed before shedding of their seeds to reduce the spread of weeds.

Cultural control:

Potato tuber moth

- After harvesting, potatoes should be kept in heaps in cool places for another 10-15 days for drying and further curing of skin. Heaps 3-4 meter long, wide at the base and about 1 meter wide at the top are the best. In hills the harvested potatoes are spread in well-ventilated rooms for drying.
- Fresh market potatoes should be stored between 5 to 6 °C. Potatoes that are used for making chips should be stored between 7 and 10 °C.
- Collection of left over tubers in the field after harvesting
- Storage of healthy tubers in moth proof structures

Bacterial soft rot

Cultural control:

- Physical damage must be avoided as it encourages post-harvest rots.
- Before storage curing is effective and non-chemical control method can be done by exposing tubers for 5 days at 15 – 20 °C and 90 - 95% RH.

Black heart

Cultural control:

- Maintain cold storage at 4 °C or slightly higher.
- Maintain proper aeration in the storage.

Note: Pesticides dosages and spray fluid volume are based on high volume sprayer.

**Apply Trichoderma viride/harzianum and Pseudomonas fluorescens for treatment of seed tubers and soil application (if commercial products are used,*

CHECK

for label claim. However, biopesticides produced by farmers for own consumption in their fields, registration is not required).

**** Pests of regional significance**

Source: [NIPHM](#), and [Directorate of Plant Protection, Quarantine & Storage](#)

Source: <https://data.vikaspedia.in/short/lc?k=kilVZY6B1JbVDIZK-ji2uA>

